

S9 Table. Notation-to-code mapping.

Spatial modeling of tissues for morphogenic field analysis

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Each row gives one mathematical object from the manuscript, its meaning, the name it carries in the analysis code, and the script and line at which it is constructed. Line numbers refer to the versions of the scripts archived with this submission and available at <https://github.com/aaron-oz/a-s-omics>.

All paths are relative to `Supplements for Publication/code/spatial-modeling/`. The directory `supp-4-lik-comparison/` is the canonical pipeline: it produces the per-feature fitted fields used in every reported analysis, together with Fig 4 and Table 1. Within it, the three scripts referenced below run in the order 00, 01, 02, 03, and are abbreviated here as 00-launch, 01-params, 02-run and 03-post.

One caution for readers of the code. The file-level default at 02-run:32 sets the marginal standard deviation component of the Matérn penalized-complexity prior to 1. That default is overridden inside the Poisson branch at 02-run:63, which sets it to 0.1. The values reported in the manuscript, and given in this table, are the line-63 values, because the Poisson branch is what produced every reported result. Separately, `matern.pri.feats.present` at 02-run:33 and 02-run:231 belongs to the binomial (zero) component of the ZAP model and not to the Poisson model.

Symbol	Meaning	Code object	Script and line
Spatial domain and observation units			
s	A spatial location in the field of view	<code>cbind(x, y)</code>	02-run:71
S	The continuous spatial domain, here the analyzed field of view: x from 15000 to 25000 and y from 11500 to 18000 in raw chip coordinates	<code>sub.bound</code>	01-params:43
—	Bin size used throughout (bin 50)	<code>agg.size <- 50</code>	01-params:39
—	The curated feature list (918 genes) from which the modeled features are drawn	<code>to.use\$gene</code> , loaded from <code>features.to.use.2025-04-04.Robj</code>	01-params:79--80
Data stage			
$y_i(s)$	Observed count of feature i at bin s	<code>feat.count</code> , a column of <code>run.dat</code>	assembled in 00-launch; used at 02-run:76
$F_i(s)$	The count of feature i at s as a random variable; the response in the observation model	response side of <code>formula = feat.count ~ feat.count.int + feat.count.field</code>	02-run:76
$N(s)$	Observed total counts (nUMI) at s , entering as a fixed, known offset. This is the quantity the reported analyses use in place of a latent total-count model	<code>total.count</code> , passed as the exposure argument <code>E =</code>	02-run:77

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Symbol	Meaning	Code object	Script and line
—	The Poisson observation model itself	<code>poi.lik <- bru_obs(family = "poisson", ...)</code>	02-run:73--78
Process stage			
$\lambda_i(\mathbf{s})$	Modeled rate for feature i : expected counts per unit of total counts	<code>lambda <- exp(feats.count.int + feats.count.field)</code>	02-run:105
$N(\mathbf{s})\lambda_i(\mathbf{s})$	Posterior predictive mean count	<code>expect <- lambda * total.count</code>	02-run:106
α_i	Feature-specific scalar intercept	<code>feats.count.int(1)</code>	02-run:70
$u_i(\mathbf{s})$	Zero-mean spatial Gaussian process for feature i , carrying all spatial variation in $\log \lambda_i$	<code>feats.count.field(cbind(x, y), model = matern.feats.count)</code>	02-run:71
Covariance, its parameters, and their priors			
$\Sigma(\boldsymbol{\theta}_i)$	Matérn covariance of u_i , represented through the SPDE approach as a sparse precision matrix on mesh nodes	<code>matern.feats.count <- inla.spde2.pcmatern(...)</code>	02-run:64--67
$\boldsymbol{\theta}_i = (\rho_i, \sigma_i)$	Range and marginal standard deviation of the Matérn field for feature i	<code>matern.pri.feats.count <- c(500, .95, .1, .05)</code>	02-run:63
—	Range prior, $P(\rho_i < 500) = 0.95$, with range in raw chip coordinate units	<code>prior.range = matern.pri.feats.count[1:2]</code>	02-run:66
—	Marginal standard deviation prior, $P(\sigma_i > 0.1) = 0.05$	<code>prior.sigma = matern.pri.feats.count[3:4]</code>	02-run:67
ν	Matérn smoothness, fixed rather than estimated. SPDE order $\alpha = 2$ gives $\nu = 1$ in two dimensions	<code>alpha = 2</code>	02-run:64
—	Sum-to-zero constraint on the field, for identifiability against the intercept	<code>constr = TRUE</code>	02-run:65
—	SPDE triangulation of $\mathcal{S}$; approximately 10,200 nodes over the 26,000 observation bins	<code>mesh <- fm_mesh_2d(...)</code> , with <code>max.edge</code> , <code>offset</code> and <code>cutoff</code> set at lines 45–46	00-launch:48--54
The latent total-count model, which the reported analyses do NOT use			

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Symbol	Meaning	Code object	Script and line
$\lambda_N(\mathbf{s})$, α_N , $\boldsymbol{\theta}_N$, $\Sigma(\boldsymbol{\theta}_N)$	Objects of the latent model for the total counts. These appear in the general model specification in the manuscript but are not used in any reported result. In the canonical pipeline the corresponding code is commented out; the live implementation is in the Supplement 3 pipeline	<code>matern.pri.total</code> , <code>matern.remain</code> (commented out)	commented at 02-run:31 and 02-run:35--38; live at supp-3-raw-v-norm-joint-feat-total-model/02-
Zero-inflated and zero-adjusted variants			
$p_i(\mathbf{s})$ (ZIP)	Spatially varying probability of a zero observation under the zero-inflated Poisson	<code>zip.lik <- bru_obs(...)</code>	02-run:143--163
$p_i(\mathbf{s})$ (ZAP)	Spatially varying zero probability under the zero-adjusted Poisson, with its own Matérn field and prior <code>c(1000, .95, 10, .05)</code>	<code>feat.present.int</code> , <code>feat.present.field</code> , <code>matern.feat.present</code> , <code>zap.feat.lik</code>	02-run:231--253
Posterior summaries and prediction scores			
—	Number of posterior samples drawn per feature	<code>n.samp <- 1000</code>	02-run:55
$AE_i(\mathbf{s}_j)$	Absolute error. Note that this one score uses the posterior predictive <i>median</i> , whereas <i>SE</i> and <i>DS</i> use the mean	<code>AE = abs(count - pred.median)</code>	02-run:385
$SE_i(\mathbf{s}_j)$	Squared error	<code>SE = (count - pred.mean)^2</code>	02-run:386
$DS_i(\mathbf{s}_j)$	Dawid-Sebastiani score	<code>DS = (count - pred.mean)^2 / pred.var + log(pred.var)</code>	02-run:387
$LS_i(\mathbf{s}_j)$	Negated logarithmic score	<code>LG = log_score</code> , built as <code>-log(pred.poi\$obs_prob\$mean)</code>	02-run:388; constructed at 02-run:129
MAE_i , $RMSE_i$, MDS_i , MLG_i	The four per-feature scores of Table 1, each averaging its location-level score over observation locations. Written to <code>{feature}-scores.csv</code>	<code>MAE = mean(AE)</code> , <code>RMSE = sqrt(mean(SE))</code> , <code>MDS = mean(DS)</code> , <code>MLG = mean(LG)</code>	02-run:394--397; written at 02-run:750
—	Table 1, absolute averages across features	<code>sum.res</code>	03-post:69--84

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Symbol	Meaning	Code object	Script and line
—	Table 1, parenthetical percentages. Each is a per-feature ratio to the Poisson score, formed before averaging	<code>rel[2, 3:6] <- rel[2, 3:6] / rel[1, 3:6]</code> , summarized as <code>sum.rel</code>	03-post:58--59 and 03-post:77